25. Perform Sharpening of Image using Gradient masking.

−1	−2	−1
0	0	0
1	2	1

−1	0	1
−2	0	2
−1	0	1

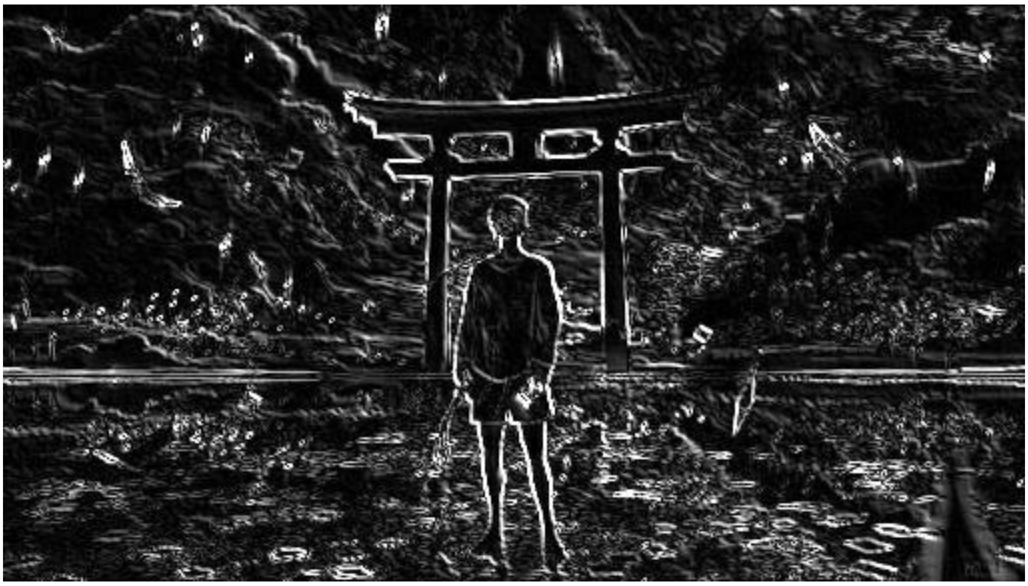
AIM:
Sharpening of Image using Gradient masking.

PROGRAM:

```
import cv2 import
numpy as np

image_path = r"C:\Users\Susmitha\Downloads\ITA0510 Lab\exp1.png"

image = cv2.imread(image_path)

if image is None:
   print("Error: Image not found!") else:
   # Convert image to grayscale     gray =
cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

   # Compute Sobel gradients in X and Y directions     gradient_x =
cv2.Sobel(gray, ddepth=cv2.CV_64F, dx=1, dy=0, ksize=3)     gradient_y =
cv2.Sobel(gray, ddepth=cv2.CV_64F, dx=0, dy=1, ksize=3)

   # Calculate the gradient image     gradient =
cv2.subtract(gradient_x, gradient_y)

   # Convert to 8-bit for proper display and saving
gradient = cv2.convertScaleAbs(gradient)

   # Save the output image     cv2.imwrite(
r"C:\Users\Susmitha\Downloads\ITA0510 Lab\sharpened_image3.jpg",
gradient
   )

   # Display the images
cv2.imshow("Original Image", image)
cv2.imshow("Gradient Image", gradient)

   cv2.waitKey(0)
cv2.destroyAllWindows()
```

INPUT:

OUTPUT:



RESULT:
Thus the program is executed successfully.